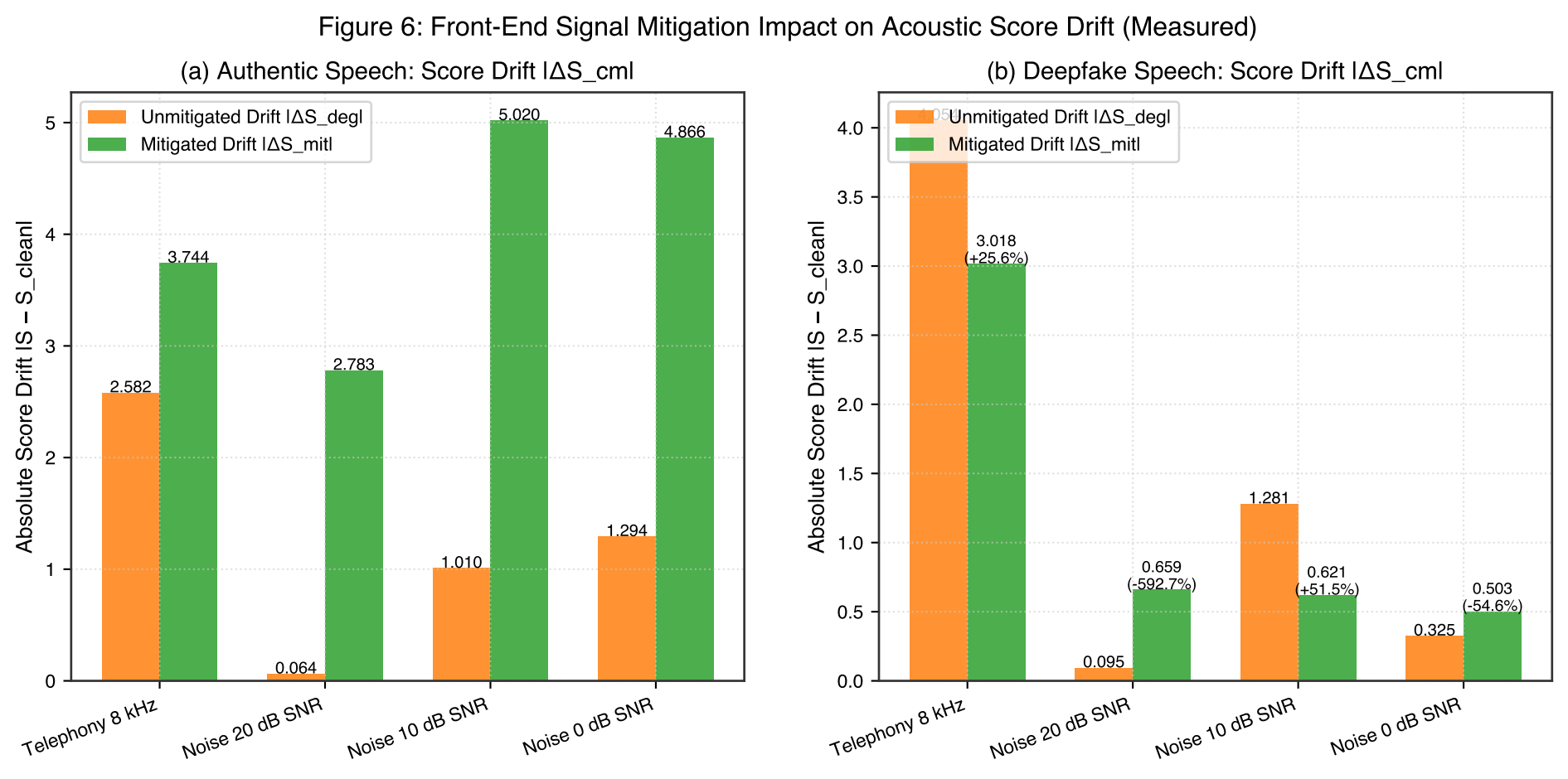
Figure 6: Front-End Signal Mitigation Impact on Acoustic Score Drift (Measured)
(a) Authentic Speech: Score Drift |ΔS_cml
(b) Deepfake Speech: Score Drift |ΔS_cml
Unmitigated Drift |ΔS_degl
Mitigated Drift |ΔS_mitl
Absolute Score Drift |S – S_cleanl
2.582
3.744
0.064
2.783
1.010
5.020
1.294
4.866
3.018
(+25.6%)
0.095
0.659
(-592.7%)
1.281
0.621
(+51.5%)
0.325
0.503
(-54.6%)
Telephony 8 kHz
Noise 20 dB SNR
Noise 10 dB SNR
Noise 0 dB SNR